

ACRE 2026 Storyboard

Purpose: a concise storyline/storyboard for validating activities, communities, products and references before development of the final presentation for the presentation : **“From observations to Climate Services: The Journey of One Observation”**

Sequence 1 — The Observation

Main Message

A historical weather observation can outlive the event and the purpose for which it was originally made.

The storm passed. The observation remained.

SC-CLI / ET Contribution

- No specific SC-CLI action is introduced at this stage.
- The observation is the starting point of a journey that will later involve preservation, Data Rescue, stewardship, climate information, services and action.

Links with Other Communities

- Historical observers and observing networks
- Archives and data holders
- ACRE and the wider Data Rescue community
- WMO Members and climate-data communities

Visual Concept

Begin with authentic historical material: a coastal semaphore or observing site, a storm warning and a handwritten observation.

Introduce the observation before SC-CLI or WMO.

Keep the historical document as the visual anchor.

WMO References

- Historical documents should be referenced on the slide, with archival source and date where available.

Possible additions / corrections

Are any important aspects of the historical observation or its context missing?

Sequence 2 — Who are we?

Main Message

The story is told mainly from the perspective of the WMO Standing Committee on Climate Services (SC-CLI), a community of worldwide voluntary experts contributing to different parts of the climate services journey.

SC-CLI supports Members across the climate services value cycle, from climate data and information to services, capacity development and informed action.

The Expert Teams reflect the WMO regional structure, while WMO also seeks an appropriate gender balance among its experts.

SC-CLI / ET Contribution

- ET-DDS — Making climate data trustworthy, accessible and reusable.
- ET-MCCVC — Producing reliable and authoritative climate information on variability, change and extremes.
- ET-CIDS — Connecting climate information with user needs and supporting its transformation into usable climate services.
- ET-CDC — Strengthening the capacities that enable Members to develop, deliver and sustain climate services.

Links with Other Communities

- WMO Members and other WMO bodies
- Research and scientific communities
- Data and reanalysis communities
- ACRE and other historical-data communities
- Users of climate information and services

Visual Concept

Show WMO — Standing Committee on Climate Services, with the four Expert Teams around it.

The visual should convey a worldwide community of voluntary experts, not an administrative structure.

WMO References

No additional references are required here; relevant sources are consolidated in Sequence 11.

Possible additions / corrections

Are any important aspects of SC-CLI, its Expert Teams, their representativeness or its wider community missing or inaccurately represented?

Sequence 3 — Preserve & Make Available

Main Message

How can an observation survive beyond its original purpose?

Historical observations need to be preserved, rescued and made available so that they can contribute to future climate knowledge.

Preserving the observing heritage is the first step; rescuing the records gives them a second life.

An observation preserved becomes part of our climate memory.

But once recovered, can we trust the resulting data?

SC-CLI / ET Contribution

- Preserve Centennial and Long-Term Observing Stations and their observing heritage.
- Support Members in developing systematic Data Rescue strategies.
- Provide guidance for rescuing and digitizing historical climate records across land, marine and hydrological domains.
- Facilitate access to rescued records through the WMO Data Rescue Portal, in collaboration with Copernicus.
- WMO-No. 1238 provides a framework for Members and their Entities to develop and coordinate Data Rescue activities.

Links with Other Communities

- ACRE and other Data Rescue communities
- WMO Members and NMHSs
- Archives and other data holders
- Universities and research institutions
- Volunteers and other initiatives
- Copernicus / C3S

Visual Concept

Historical observation → archive → rescued/digitized record.

Introduce Centennial / Long-Term Observing Stations, Data Rescue and the WMO Data Rescue Portal.

Keep the historical documents as the visual anchor.

WMO References

- WMO-No. 1238, Manual on the High-Quality Global Data Management Framework for Climate
- WMO-No. 1182, WMO Guidelines on Good Practices for Data Rescue — land, marine and hydrological observations
- WMO-No. 1354, Guidelines on the WMO Recognition of Long-Term Observing Stations
- WMO framework for Centennial and Long-Term Observing Stations
- WMO Data Rescue Portal / WMO–Copernicus collaboration

Possible additions / corrections

Are any important Data Rescue activities, communities, WMO mechanisms or references missing or inaccurately represented?

Sequence 4 — Build Trustworthy Data

Main Message

Recovering an observation is not enough. It needs to become trustworthy, well-documented and reusable climate data.

SC-CLI / ET Contribution

SC-CLI/WMO contributes guidance, specifications and frameworks that support these practices.

- Make data trustworthy through documentation, metadata and provenance; quality control and quality assurance; assessment; homogenization; data stewardship; and appropriate data exchange.
- Focus on understanding and documenting the history of the data rather than simply correcting the original observation.

Links with Other Communities

- WMO Members and NMHSs
- Climate data managers and stewards
- Observation and research communities
- Homogenization and climate-data experts
- Data users and other communities relying on climate datasets

Visual Concept

Historical record → Metadata & provenance → Quality / Assessment → Homogenization where appropriate → Stewardship → Trustworthy, well-documented and reusable climate data

Show these as layers around the data rather than as a mandatory linear processing chain.

WMO References

- WMO-No. 100, Guide to Climatological Practices
- WMO-No. 1131, Climate Data Management System Specifications
- WMO-No. 1269, Guidelines on Surface Station Data Quality Control and Quality Assurance for Climate Applications
- WMO-No. 1245, Guidelines on Homogenization
- WMO-No. 1328, Stewardship Maturity Matrix for Climate Data (SMM-CD)
- DAYCLI Guidelines

Possible additions / corrections

Are any important SC-CLI activities, communities, practices or WMO references missing or inaccurately represented?

Sequence 5 — From Data to Climate Information

Main Message

Trustworthy datasets allow us to describe, monitor and understand the climate.

The observation has become evidence.

But these datasets do not exist in isolation. What happens when observations, datasets and models are brought together?

SC-CLI / ET Contribution

- Turn trustworthy climate datasets into climate information through climate normals, key climate indicators, climate indices, assessment of extremes and climate monitoring.
- Support assessment and stewardship of datasets through the SMM-CD and their discoverability through the WMO Data Catalogue.
- Include new and emerging indicators as climate questions evolve.

Links with Other Communities

- WMO Members and NMHSs
- Climate monitoring and assessment communities
- Research and scientific communities
- Data and modelling communities
- ACRE and other observational communities
- Users of climate information

Visual Concept

Trustworthy climate datasets → Climate Normals | Climate Indices | Key Climate Indicators | Extremes | Climate Monitoring.

Keep the historical station observations visually connected to the resulting climate information.

The SMM-CD and WMO Data Catalogue can appear as supporting elements rather than as separate steps.

WMO References

- WMO-No. 100, Guide to Climatological Practices
- WMO-No. 1238, Manual on the High-Quality Global Data Management Framework for Climate
- WMO-No. 1328, Stewardship Maturity Matrix for Climate Data (SMM-CD)
- WMO-No. 1203, WMO Guidelines on the Calculation of Climate Normals
- WMO-No. 847, Climatological standard normals (CLINO) for the period 1991–2020
- WMO/TD-No. 1377, WCDMP-61, The Role of Climatological Normals in a Changing Climate
- WMO-No. 1186, Guidelines for the Submission of the World Weather Records 2011+
- WMO-No. 1173, Handbook of Drought Indicators and Indices
- WMO-No. 1310, Guidelines on the Definition and Characterization of Extreme Weather and Climate Events
- WMO-No. 1287, Developing the Climate Science Information for Climate Action
- WMO Data Catalogue
- ET-CCDI and Climdex websites
- WMO Key Climate Indicators website

Possible additions / corrections

Are any important climate products, SC-CLI activities, communities or WMO references missing or inaccurately represented?

Sequence 6 — From Observational Datasets to the Bigger Climate Picture

Main Message

One observation contributes to a much bigger picture — across datasets, methods and timescales.

Observations, datasets and modelling systems contribute to climate information and climate assessments at different spatial and temporal scales.

SC-CLI / ET Contribution

- Connect observations and climate datasets with monitoring, prediction, projection, State of the Climate reporting and climate assessments.
- Recognize reanalysis as one type of gridded climate product produced using observations together with modelling systems.
- Use long, quality-assured climate records to assess and validate gridded climate products and support climate assessments.
- SC-CLI contributes to this broader information chain but does not manage the whole production landscape.

Links with Other Communities

- IPCC and climate assessment communities
- ACRE and historical Data Rescue communities
- WMO Members and observational networks
- Climate-data and research communities
- NWP, modelling and reanalysis communities
- WIPPS and operational prediction communities
- Regional Climate Centres and seasonal prediction initiatives
- CORDEX and regional downscaling communities
- Copernicus and other international climate-data initiatives

Visual Concept

Expand from trustworthy datasets to: Many observations + Many datasets + Models / NWP + Other sources.

Then: Gridded climate products — Reanalysis | Prediction | Projection — and State of the Climate reports that bring climate information together into a coherent picture.

Show information expanding from Global → Regional → National / Local.

Include seasonal prediction, regional datasets, downscaled prediction and downscaled projections.

Show a feedback link from long climate records to Assessment / Validation of gridded products.

Keep the original historical observation visible within the wider network.

A possible case study is the development of global temperature datasets from long observational records, illustrating how observations contribute both to climate products and to climate assessments.

WMO References

- WIPPS — WMO Integrated Processing and Prediction System
- WMO/global and regional seasonal prediction initiatives
- Regional Climate Centres and their products and datasets
- Relevant WMO frameworks for regional and national downscaling of prediction and projection information
- Relevant WMO and partner frameworks for reanalysis and gridded climate products
- CORDEX and its regional climate downscaling datasets and products
- CSIS and its data, monitoring, prediction and projection functions
- Copernicus: European State of the Climate — Report 2025
- WMO-No. 1395, State of the Climate in Latin America and the Caribbean 2025
- WMO-No. 1400, State of the Climate in the South-West Pacific 2025
- WMO-No. 1398, State of the Climate for Africa
- WMO-No. 1397, State of the Climate in Asia 2025
- Regional State of the Climate reports
- WMO-No. 1391, State of the Global Climate 2025
- IPCC Assessment Reports and relevant climate assessment products

Possible additions / corrections

Are any important datasets, methods, initiatives, communities, SC-CLI activities or WMO references missing or inaccurately represented?

Sequence 7 — From Climate Information to Climate Services

Main Message

Climate information becomes a climate service when it responds to users' needs and is supported by appropriate operational systems and dialogue.

Can every Member actually deliver and sustain these services?

SC-CLI / ET Contribution

SC-CLI/WMO helps Members organize, tailor and deliver climate information through operational systems and dialogue with users.

- Standardize — common approaches for generating, exchanging and delivering climate information products.
- Tailor — user-targeted and sector-specific information, including climate risk management and climate services / indices.
- Enable — operational CSIS from global to regional and national scales, including NFCS and GPC–RCC–NMHS linkages.
- Coordinate / Collaborate / Co-produce — work across GFCS, RFCS and NFCS levels, RCCs, RCOFs/NCOFs, WIPPS, INFCOM, the Research Board and partners.
- Facilitate — user–provider dialogue, multidisciplinary interaction and co-development with sectoral users.

Links with Other Communities

- WMO Members and NMHSs
- GPCs and RCCs
- RCOFs / NCOFs and regional and national climate forums
- WIPPS and operational prediction communities
- INFCOM and research communities
- Sectoral users and other partners
- GFCS / RFCS / NFCS communities
- Copernicus and other international initiatives

Visual Concept

Climate information → tailored information → climate services.

Show a two-way relationship between providers and users.

GPCs → RCCs → NMHSs → Users, with NFCS / CSIS as supporting structures.

Keep the emphasis on dialogue and co-production, not on a one-way delivery.

WMO References

- CSIS Technical Reference
- Climate Services Toolkit
- Guidelines on CSIS Regional Implementation
- Guidelines on Operational Sub-seasonal Prediction
- Guidelines for RCOF/RCF Operations
- Guidelines for NCOF/NCF Operations
- Guidelines on the Interpretation and Use of Regional Climate Change Projections
- Guidelines on the Establishment of National Frameworks for Climate Services (NFCS)
- Guidelines on Climate Risk Management
- Guidelines on Tailoring Climate Information
- Guidelines for the Establishment and Operation of RCCs — Designation and Assessment
- Guidance for National CSIS Focal Points for Climate Services
- Major technical contributions to be checked: Operational CSIS; RCC/nRCC networks and assessment; Operational Climate Prediction workshops; WMO ENSO mechanism; RCOF/RCF and NCOF/NCF operations and training; Climate Services Toolkit and Clim pact; ClimSA User Interface Platform; State of Climate Services Report; EW4All; NFCS; and integration with INFCOM/WIPPS and WCRP RfS.

Possible additions / corrections

Are any important ET-CIDS activities, partner communities, user groups or WMO references missing or inaccurately represented?

Sequence 8 — Enabling Members to Deliver and Sustain Climate Services

Main Message

Climate services become sustainable when people and institutions have the capacity to deliver them.

But why are we doing all this? To support action.

SC-CLI / ET Contribution

- Develop → Train → Assess → Strengthen.
- Competency-based training and learning resources, including digital learning.
- Quality management approaches and maturity assessment of climate-service capacity.
- Implementation guidance and support to NMHSs.
- Help Members move from 'A climate service exists' to 'A Member can actually deliver and sustain it.'

Links with Other Communities

- WMO Members and NMHSs
- RCCs and other regional structures
- Universities and training institutions
- Climate-service experts and practitioners
- Other WMO communities and development / international partners

Visual Concept

Institution → People → Skills → Practice → Operational service.

Possible elements: NMHSs, national teams, learning, tools and guidance, assessment, implementation.

Keep the emphasis on people and institutions.

WMO References

- WMO-No. 1247, Capacity Development Guidelines
- Basic Instructional Package for Climate Services (BIP-CS)
- Climate Services Implementation Checklist
- ET-CDC framework for assessing NMHS climate-service capacity
- Relevant quality-management / ISO 9001 guidance
- Relevant WMO digital-learning and training resources

Possible additions / corrections

Are any important ET-CDC activities, capacity-building mechanisms, Member needs, communities or WMO references missing or inaccurately represented?

Sequence 9 — From Climate Services to Decision and Action

Main Message

The value of climate information is realized when it helps people and institutions make better-informed decisions.

The journey does not end with information. It ends with action.

SC-CLI / ET Contribution

- No new Expert Team is introduced here.
- The sequence shows the collective result: Data → Information → Services → Capacity → Action.

Links with Other Communities

- Health
- Agriculture and food security
- Water
- Energy
- Disaster Risk Reduction and anticipatory action
- Cities and urban planning
- Climate adaptation and resilience
- Early Warnings for All (EW4All)

Visual Concept

Climate Information → Climate Services → Users & Decision-makers → Decisions → ACTION.

Keep the visual simple and give maximum weight to ACTION.

A discreet link to the WMO Climate Services Dashboard may support the transition, without introducing it as a new technical object.

WMO References

No additional references are required here; relevant sources are consolidated in Sequence 11.

Possible additions / corrections

Are any important areas of climate action, user communities or SC-CLI contributions missing or inaccurately represented?

Sequence 10 — Back to the Observation: Completing the Climate Services Value Cycle

Main Message

One observation. A much bigger journey.

After following the journey from a historical observation through data, information, climate services and action, we return to where the journey began.

The journey continues as new needs and decisions shape future climate services, information, data and observations.

The journey continues.

SC-CLI / ET Contribution

- No new SC-CLI activity is introduced.
- The four Expert Teams have contributed to different, interconnected parts of the Journey.

Links with Other Communities

- ACRE
- WMO Members and observing networks
- Archives and research communities
- Copernicus
- Sectoral communities and users

Visual Concept

Return to the same historical semaphore / handwritten observation used in Sequence 1.

Briefly recall: Preserve → Rescue → Quality and Stewardship → Data → Information → Services → Action.

Let the visual return to the original observation.

WMO References

No additional references are required here; relevant sources are consolidated in Sequence 11.

Possible additions / corrections

Are any important communities, partners or elements of the Journey missing from this final perspective?

Closing Message

No single community can complete the journey alone.

The journey continues — with all of us.

Sequence 11 — References

Purpose

A concise reference section will accompany the presentation, bringing together the WMO publications, guidelines, tools, datasets, initiatives and other relevant resources mentioned throughout the Journey.

- Explore the sources behind the different steps of the Journey.
- Consult relevant WMO and partner resources.
- Go further into specific topics or activities.

The reference material will be provided outside the main presentation, through a link and/or supporting slides.

PRESENTATION NOTE — NOT PART OF THE STORYBOARD

Elements to be conveyed visually or through the speaker rather than added to the storyboard text.

- S5 → S8 — Assessment
- S5: Assess the data
- S8: Assess the capacity
- S8 — Return to the human dimension
- S8 marks the return to the human dimension of the Journey: people, institutions, skills and operational capacity.
- S9 — Climax of the Journey: keep this sequence very short and visually simple. The final emphasis must be on ACTION. Do not overload it with lists of services, programmes or organisations.
- Transition to S10: after ACTION, use a short pause, then “So where did our journey begin?” before returning to the original observation.
- **S10 — Closing loop:** the return to the original observation should show that the audience now sees the same historical record differently: not as an isolated old document, but as one element of a much larger system of knowledge, collaboration and climate action.
- **Feedback loop:** decisions and user needs can generate new requirements for climate services, information, data and observations.
- **Keep the ending open:** the Journey does not end; it continues.

Terminology Note

Trustworthy

Describes a property of climate data or datasets: they are sufficiently reliable, well-documented, quality-assured and fit for their intended use.

Stewardship

Refers to the coordinated practices used to manage climate data throughout their lifecycle, including quality, integrity, documentation, metadata, accessibility, processing, preservation and long-term usability.

Trust

Describes a relationship between users and data or datasets — whether users have confidence in the data and the reasons for that confidence.

This distinction is useful for the storyboard because “trustworthy” describes the data, while “trust” describes the relationship with its users; “stewardship” describes the practices that help make and keep data trustworthy.